AP PRECALCULUS

Unit 1 • Lesson 1.7 practice

Lesson 1.7 • Rational Functions and End Behavior

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the horizontal asymptote of $r(x) = \dfrac{5x-1}{x^2+4}$?

(A) $y = 5$
(B) $y = 0$
(C) $y = -\dfrac{1}{4}$
(D) There is none.

Question 2

NO CALCULATOR

What is the horizontal asymptote of $r(x) = \dfrac{4x^3 - x}{1-2x^3}$?

(A) $y = 2$
(B) $y = 4$
(C) $y = -2$
(D) $y = 0$

Question 3

NO CALCULATOR

What is the slant asymptote of $r(x) = \dfrac{x^2+3x}{x-1}$?

(A) $y = x+3$
(B) $y = x+4$
(C) $y = x$
(D) $y = x-4$

Question 4

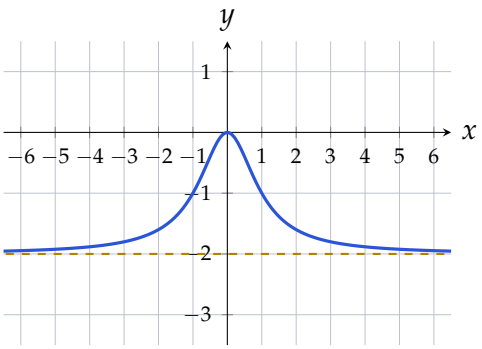
NO CALCULATOR

$\lim\limits_{x\to-\infty} \dfrac{x^3}{x^2+1}$ is

(A) 1
(B) 0
(C) $+\infty$
(D) $-\infty$

Question 5 (graph) NO CALCULATOR

The graph of $r(x) = \dfrac{-2x^2}{x^2+1}$ is shown. What is $\lim\limits_{x\to\infty} r(x)$?



(A) 0
(B) -2
(C) $-\infty$
(D) 2

Question 6 (Free response) NO CALCULATOR

Let $r(x) = \dfrac{6x^2+x}{3x^2-12}$ and $s(x) = \dfrac{2x^2-x+3}{x+1}$.

(a) Write limit statements for the end behavior of r.
(b) Find the equation of the slant asymptote of s.
(c) For large positive values of x, is the graph of s above or below its slant asymptote? Explain.